Korokoro Environmental Group

Water quality & biomonitoring summary 2026

Korokoro Environmental Group








Greater Wellington
Te Pane Matua Taiao

Front cover photos – left to right:

Korokoro Environmental Group open day event. Image © Greater Wellington

Shyam spotlight fishing. Image © Greater Wellington

Bluegill bully. Image © EOS Ecology

Local Playcentre visit. Image © Greater Wellington

Smooth-cased caddis *Olinga*. Image © EOS Ecology

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Korokoro Environmental Group (KEG) event – kids assessing MCI. Image © Greater Wellington

1. Korokoro Environmental Group (KEG) overview

KEG is an incorporated society based in the Western Hills of the Hutt Valley. Our group has a long history of advocating for the preservation of our natural environment and in recent years, have increasingly turned our attention to the Korokoro Stream.

Korokoro Environmental Group

keg.org.nz

KEG is an incorporated society based in the Western Hills of the Hutt Valley. Our group has a long history of advocating for the preservation of our natural environment and in recent years, have increasingly turned our attention to the Korokoro Stream. The stream, the only fully vegetated stream entering into Wellington Harbour, is a taonga for mana whenua, our community, many day-trippers that visit from across the region, as well as the thousands of other non-human species that call it home and are recorded in this report.

By cataloguing the stream's condition, we hope to educate, strengthen our connection with the stream and other kaitiaki, and provide a resource to assist in future stream protection.

Monitoring outlined in this report has been conducted across seven visits – two measuring eDNA, four looking at water quality indicators and one monitoring fish populations. Samplers have come from our community: Tessa Roberts, Lesley Smith, Tamsin Falconer, supported by Anthony Davies, Nicola McCarthy, Gillian Woodward, and Andre Oliver, with resourcing and expertise provided by Amanda Valois of Greater Wellington (GW).

The outputs of monitoring catalogued in this report describe only one small part of what stream monitoring has taught us. The practice of regularly visiting and observing has revealed where the best blackberries and watercress can be found, the pool where you can always find trout, where predators attack, how the elements shift the character of the stream, and where those who are less fortunate in our community find sanctuary rough sleeping on its banks. We've also heard stories from the last 50+ years of locals visiting the stream.

A byproduct of doing this work is growing a network of other kaitiaki who also care for the stream. We have spoken with Tenths Trust, and have been heartened by their support, especially from Te Karanga O Te Tui Marino. Mountains to Sea Wellington (MTSW), especially Sam Kearns has provided us direction, and confidence to take action. Several GW staff have supported us with knowledge and resources – as well as Amanda we have been especially supported by Peta McMillan (park ranger) and Amy Gault (fish passage).

Already, this work has catalysed action. We have extended our planting sites in the stream floor, targeting plantings guided for this key native ecosystem, and advocated for revegetation in the upper reaches of the catchment. We have hosted two public open days at the stream in partnership with GW and MTSW offering a chance to explore the species and pressures on the stream. We have participated in three spot lighting events, with the community, scouts and fish passage experts. We've helped catalyse the installation of a rock ramp being installed by GW, who are working to restore passage in the stream's lower reaches.

As pressure on our environment and overburdened planet mount, Korokoro Stream provides an oasis of abundance and beauty, sited remarkably close to our urban centres and loved by many. We hope that this report will serve as a resource for others wishing to learn about the stream, advocate for the stream, or planning and executing future works around the stream to support the health of this precious waterbody.



Korokoro Environmental Group (KEG) open day event. Image © Greater Wellington

2. Catchment overview

The Korokoro Stream catchment covers about 1,647 hectares and is one of the most naturally vegetated areas flowing into Wellington Harbour. Most of the trees and vegetation present today have regrown after earlier land clearance for farming, meaning much of the catchment is now covered in regenerating native forest and scrub. There is a remnant of the original forest type in the vicinity of Korokoro Dam, and much of the catchment is included in GWs Key Native Ecosystem Programme (Greater Wellington, 2020).

If left entirely undisturbed, the catchment would naturally support different types of native forest. Modelling has shown the lower catchment would be dominated by rimu-tawa-kamahi forest, while higher up rimu-broadleaf-beech forest would be prevalent (Our Environment, 2026). The land itself is steep, and the underlying rock is very old sedimentary material made up mainly of sandstone and mudstone, formed around 142–248 million years ago (GNS Science, 2020).

Looking at how the land is used today, based on the Land Cover Database (Manaaki Whenua Landcare Research, 2025 (LCDB 6.0)), nearly two-thirds (63.3%) of the catchment is covered in native broadleaf forest. Smaller areas include pasture (13.9%), exotic (non-native) forest (8.7%), and a small amount of urban or built-up land (about 4.5%), mostly in the lower, southeast of the catchment (LCDB 6.0).

GW has been monitoring the health of the stream at Cornish Street Bridge since 2022, collecting regular data on water quality and ecology. In addition to this, KEG has carried out its own monitoring, including water quality testing at nine different sites, environmental DNA (eDNA) sampling to identify species present, and spotlighting surveys of fish. This report brings together information from KEG, GW, and other sources to provide an overall picture of the stream's health.

Korokoro Environmental Group (KEG)
Focus area map:

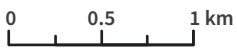
- Waterways
- Korokoro Stream catchment boundary
- Historic dams

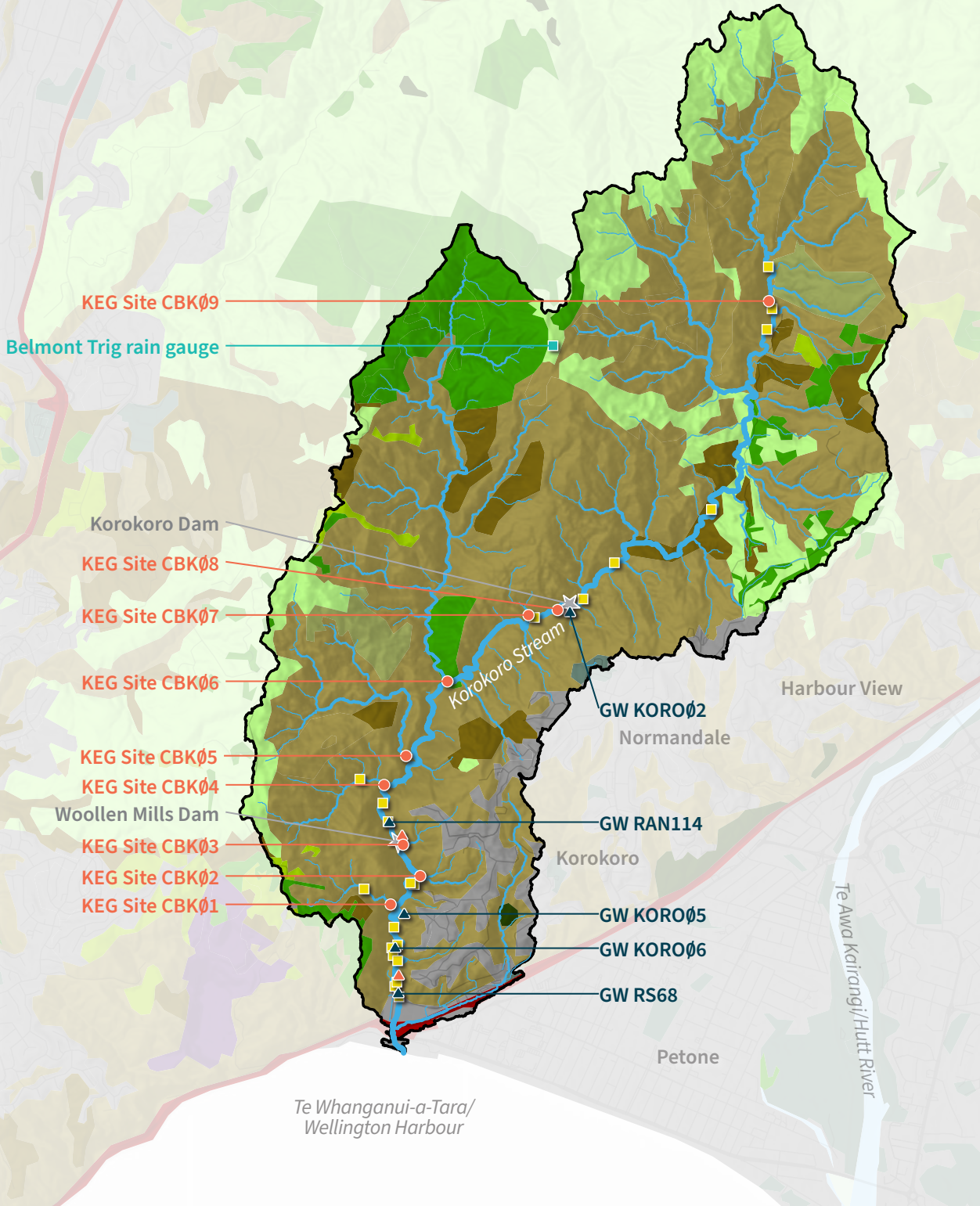
Monitoring sites:

- KEG water quality
- GW ecology
- GW rainfall
- Public eDNA
- New Zealand Freshwater Fish Database (NZFFD)

Land cover (LCDB) type & coverage:

	Catchment Area (ha)	(%)
Low Producing Grassland	11	0.67
High Producing Exotic Grassland	229.6	13.9
Gorse and/or Broom	92.9	5.6
Transport Infrastructure	4.7	0.3
Forest – Harvested	44	2.7
Exotic Forest	142.9	8.7
Built-up Area (settlement)	74.5	4.5
Broadleaved Indigenous Hardwoods	1,042.4	63.3
Indigenous Forest	1.9	0.1
Urban Parkland/Open Space	3.2	0.2





3. Existing state summary

GW started monitoring the Korokoro Stream at the Cornish Street Bridge site in July 2022. Because this is still a relatively short period of time, there isn't yet enough data to clearly identify long-term trends. The results presented here should therefore be seen as an early snapshot rather than firm conclusions.

Looking at the aquatic macroinvertebrates, the results suggest the stream is in reasonably good condition overall, although there are some signs of pressure. The key measures used are all still above the national minimum standards, which is positive (Table 1).

The water itself is generally clear, with very little sediment in suspension. However, nutrient levels are of concern. Nitrogen levels exceeded ANZG (2018) guideline values, and while they are not currently at toxic levels, they may still contribute to issues like excessive algal growth. Phosphorus shows a similar pattern: total levels are low, but the form that is easily used by algae is above recommended guidelines.

These elevated nutrients may help explain the algae growth that has been observed. Monitoring since 2022 has shown periods where algal cover has been very high, reaching up to 80% of the streambed in December 2023.

Bacteria levels (*E. coli*) are generally low, which is encouraging. However, occasional high readings mean the site is still rated as having some risk in terms of recreational use.

Overall, the stream is in good condition in many respects, but nutrients and algal growth are areas to keep an eye on.

TABLE 1: A summary of biomonitoring and water quality data from GW's Korokoro Stream at Cornish St Bridge long-term state of the environment monitoring site. This site was only established in July 2022, so is yet to have the minimum five years of data to apply NPS-FM (2020) attribute bands and calculate trends. The NPS-FM (2020) attribute states shown are indicative only and ANZG (2018) default guideline values (DGVs) are also shown where available. Where ANZG (2018) DGVs are available, median values are coloured green or red depending on whether they meet or do not meet these guideline values, respectively.

Parameter	Length of Record	Median Value	Indicative NPS-FM (2020) Attribute Band	ANZG (2018) DGVs
MCI	3 years	112.6	B	No value
QMCI		4.66	C	No value
ASPM		0.55	B	No value
E. coli	3 years, 10 months	34 n/100 mL	C	No value
Visual Clarity		3.73 m	A	1.4 m
Turbidity		0.47 NTU	Not an NPS-FM (2020) attribute	2.3 NTU
Total Nitrogen		0.5 mg/L	Not an NPS-FM (2020) attribute	0.272 mg/L
Total Oxidised Nitrogen		0.4 mg/L	Not an NPS-FM (2020) attribute	No value
Dissolved Inorganic Nitrogen		0.4 mg/L	Not an NPS-FM (2020) attribute	No value
Ammoniacal Nitrogen		<0.005 mg/L (below detection)	A (toxicity)	0.009 mg/L
Nitrate Nitrogen		0.40 mg/L	A (toxicity)	0.17 mg/L
Dissolved Reactive Phosphorus		0.015 mg/L	C	0.011 mg/L
Total Phosphorus		0.017 mg/L	Not an NPS-FM (2020) attribute	0.018 mg/L

Based on available fish records, 12 freshwater fish species have been found in the Korokoro Stream catchment (Table 2). The species most often seen are longfin eel, redfin bully, and brown trout. Some of these fish are under pressure: four species are considered At Risk – Declining, three are At Risk – Naturally Uncommon, and one species, īnanga, is classed as Threatened – Nationally Vulnerable.

Most fish surveys have taken place in the easier-to-reach lower parts of the catchment. However, surveys further upstream show that several migratory fish are still able to move past barriers such as the historic Woollen Mills Dam and Korokoro Dam to reach habitats higher up the stream. Even so, these structures may reduce the number of fish that successfully make it upstream.

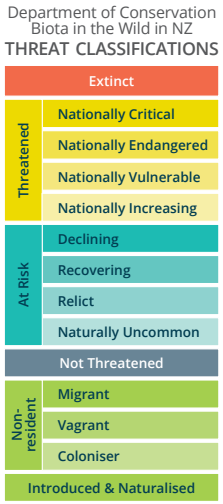
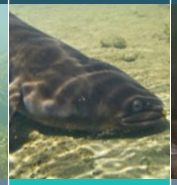
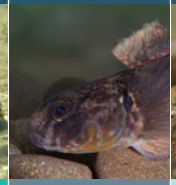
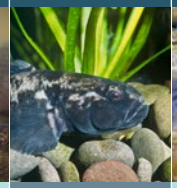
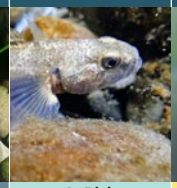
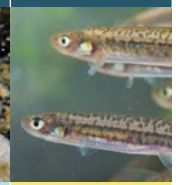
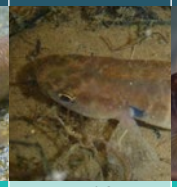
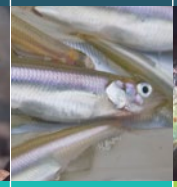
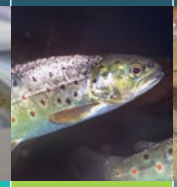


TABLE 2: A summary of New Zealand Freshwater Fish Database fish records (including kōura) and GW fish survey data from the Korokoro Stream catchment, including the years that fish surveys have found that species, relative to the location of the historic Woollen Mills Dam and Korokoro Dam. The conservation status for each fish species from Dunn *et al.* (2025) and for kōura from Grainger *et al.* (2018) is shown.

DOWNSTREAM OF WOOLLEN MILLS DAM							
	Shortfin eel *Anguilla australis*	**Longfin eel** *Anguilla dieffenbachii*	**Redfin bully** *Gobiomorphus huttoni*	**Bluegill bully** *Gobiomorphus hubbsi*	**Common bully** *Gobiomorphus cotidianus*	**Dinah's bully*** *Gobiomorphus dinae*	**Īnanga** *Galaxias maculatus*
Species & Conservation Status							
	Not Threatened	At Risk – Declining	Not Threatened	At Risk – Naturally Uncommon	Not Threatened	At Risk – Naturally Uncommon	Threatened – Nationally Vulnerable
Years Recorded	1982, 2007, 2016, 2024	1974, 1975, 1982, 2007, 2009, 2016, 2022, 2024, 2025	1974, 1975, 1982, 1985, 2007, 2009, 2016, 2022, 2024, 2025	1975, 1982, 2007, 2016, 2022, 2024	1985, 2007, 2024, 2025	2022	2007, 2009, 2016, 2024, 2025

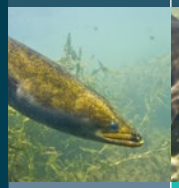
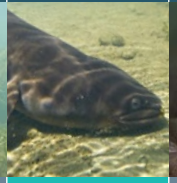
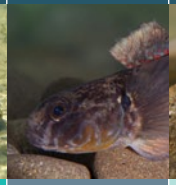
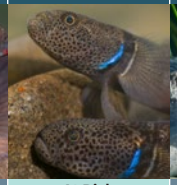
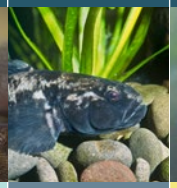
	Kōaro *Galaxias brevipinnis*	**Banded kōkopu** *Galaxias fasciatus*	**Giant kōkopu** *Galaxias argenteus*	**Common smelt** *Retropinna retropinna*	**Brown trout** *Salmo trutta*	**Kōura** *Paranephrops planifrons*
Species & Conservation Status						
	At Risk – Declining	At Risk – Naturally Uncommon	At Risk – Declining	At Risk – Declining	Introduced & Naturalised	Not Threatened
Years Recorded	2007	2009, 2016	1982, 2009, 2024, 2025	2009	1974, 1975, 1982, 1985, 2007, 2009, 2016, 2024, 2025	2016

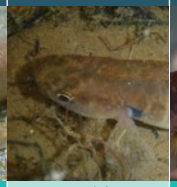
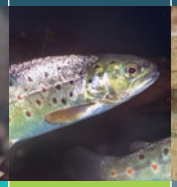
All species images © EOS Ecology unless otherwise stated.

* Previously known as Cran's bully (*Gobiomorphus basalis*)

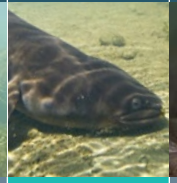
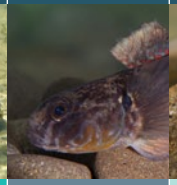
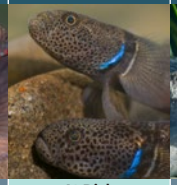
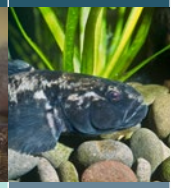
TABLE 2: ...continued...

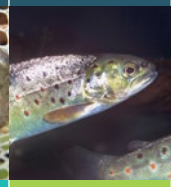
BETWEEN WOOLLEN MILLS DAM & KOROKORO DAM

	Shortfin eel *Anguilla australis*	Longfin eel *Anguilla dieffenbachii*	Redfin bully *Gobiomorphus huttoni*	Bluegill bully *Gobiomorphus hubbsi*	Common bully *Gobiomorphus cotidianus*	Īnanga *Galaxias maculatus*
Species & Conservation Status						
	Not Threatened	At Risk – Declining	Not Threatened	At Risk – Naturally Uncommon	Not Threatened	Threatened – Nationally Vulnerable
Years Recorded	2019, 2025	1952, 2016, 2019, 2025	1952, 2019, 2025	2019	1952	1952

	Kōaro *Galaxias brevipinnis*	Banded kōkopu *Galaxias fasciatus*	Giant kōkopu *Galaxias argenteus*	Brown trout *Salmo trutta*	Kōura *Paranephrops planifrons*
Species & Conservation Status					
	At Risk – Declining	At Risk – Naturally Uncommon	At Risk – Declining	Introduced & Naturalised	Not Threatened
Years Recorded	2019, 2025	2025	2025	2025	2016

UPSTREAM OF KOROKORO DAM

	Shortfin eel *Anguilla australis*	Longfin eel *Anguilla dieffenbachii*	Redfin bully *Gobiomorphus huttoni*	Bluegill bully *Gobiomorphus hubbsi*	Common bully *Gobiomorphus cotidianus*
Species & Conservation Status					
	Not Threatened	At Risk – Declining	Not Threatened	At Risk – Naturally Uncommon	Not Threatened
Years Recorded	1975, 2024	1975, 2016, 2020, 2024, 2025	2016, 2024, 2025	2016	2024

	Kōaro *Galaxias brevipinnis*	Banded kōkopu *Galaxias fasciatus*	Dwarf galaxias *Galaxias divergens*	Brown trout *Salmo trutta*
Species & Conservation Status				
	At Risk – Declining	At Risk – Naturally Uncommon	At Risk – Declining	Introduced & Naturalised
Years Recorded	2025	2016, 2025	2020	1975, 2016

All species images © EOS Ecology unless otherwise stated.

4. KEG data

Water quality:
Turbidity (NTU)

- The water in most parts of Korokoro Stream is generally very clear. During all four sampling visits, the main stem and some of the cleaner upper tributaries had low levels of turbidity, a measure of water clarity.
- However, a few side streams were consistently more turbid. These tributaries (CBKØ4, CBKØ3, and CBKØ1) had higher levels of sediment on each sampling occasion compared to other sampling sites.
- Even with these inputs, the water further downstream at the main monitoring site (Cornish Street Bridge) remained clear overall. On average, it was well within recommended guidelines over the monitoring period.
- In summary, Korokoro Stream is mostly clear and healthy in terms of turbidity, a measure of water clarity as mentioned above, although some smaller tributaries are contributing additional sediment inputs.

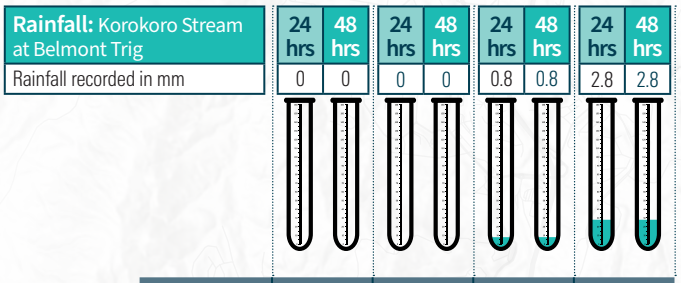
GW LONGER-TERM DATA FROM NEARBY SITES FOR COMPARISON – from www.lawa.org.nz	
GW SOE Monitoring Site	**5-year Median Turbidity**
Hutt River at Boulcott	1.32
Waiwhetu Stream at Whites Line East	4.2
Parkvale Stream at Weir	2.1
Kaiwharawhara Stream at Ngaio Gorge	1.22

Water quality: Turbidity (NTU)

ANZG (2018) DGV = 2.3
(whole catchment the same – based on Cool-Wet, Low Elevation REC climate and topography classes)

Indicates sample **OVER** DGV	Indicates sample **UNDER** DGV

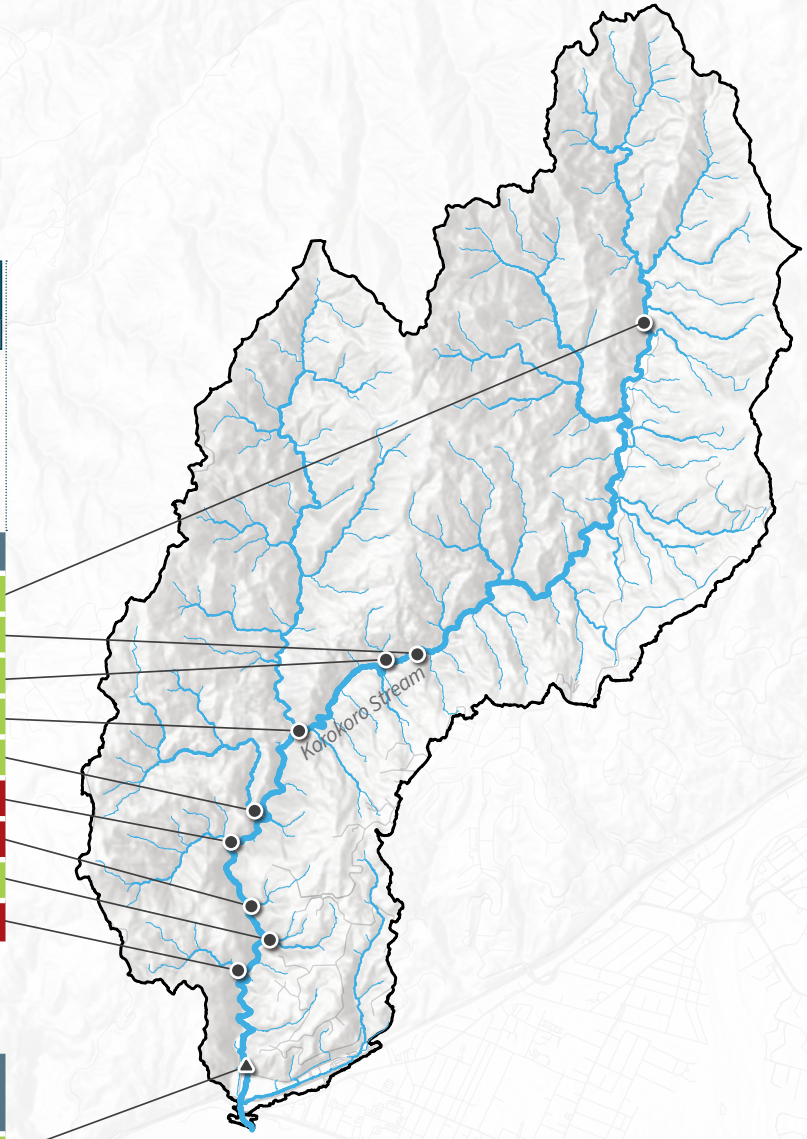
Rainfall: Korokoro Stream at Belmont Trig	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs
Rainfall recorded in mm	0	0	0	0	0.8	0.8	2.8	2.8



KEG Site #	01/07/25	21/09/25	06/11/25	15/02/26
CBKØ9	1.25	0.886	0.77	0.56
CBKØ8	2.24	1.72	0.65	0.43
CBKØ7*	0.574	0.448	0.32	1.1
CBKØ6*	2.12	0.97	0.76	0.64
CBKØ5*	2.37	1.12	0.65	1.23
CBKØ4*	3.86	3.98	3.1	2.7
CBKØ3*	4.6	4.75	3.8	4.2
CBKØ2*	2.67	1.83	5.1	1.01
CBKØ1*	6.12	4.86	2.9	3

* Indicates site on tributary of Korokoro Stream.

GW Site #	3-year 10-month Median	01/07/25	21/09/25	06/11/25	15/02/26
RS68	0.47	0.99	0.428	0.13	0.37



Water quality:
Dissolved reactive phosphorus (DRP) (mg/L)

- Levels of dissolved reactive phosphorus (DRP), a form of phosphorus that is readily used by algae, exceeded ANZG (2018) guideline values at all the sites. The only place that occasionally came in just under the guideline was the site at Cornish Street Bridge, although overall levels there were still slightly higher than recommended.
- In the main stem itself, phosphorus levels were fairly similar from upstream to downstream, with no clear pattern of them increasing or decreasing along the catchment based on the available data.
- One small side stream (CBKØ2) was notably elevated, with much higher phosphorus levels than anywhere else. This could be linked to some urban land use in that part of the catchment.
- Because most of the catchment is covered in regenerating forest rather than farmland, it's unlikely that fertilisers are the main source of this phosphorus. Instead, it's more likely coming naturally from the soils and rocks in the area.

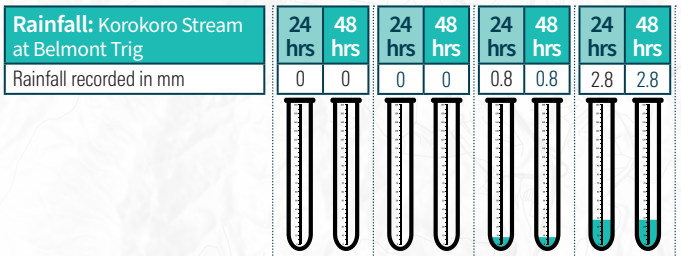
GW LONGER-TERM DATA FROM NEARBY SITES FOR COMPARISON – from www.lawa.org.nz	
GW SOE Monitoring Site	**5-year Median DRP**
Hutt River at Boulcott	0.0047
Waiwhetu Stream at Whites Line East	0.027
Parkvale Stream at Weir	0.0215
Kaiwharawhara Stream at Ngaio Gorge	0.036

Water quality: DRP (mg/L)

ANZG (2018) DGV = 0.011
(whole catchment the same – based on Cool-Wet, Low Elevation REC climate and topography classes)

Indicates sample **OVER** DGV	Indicates sample **UNDER** DGV

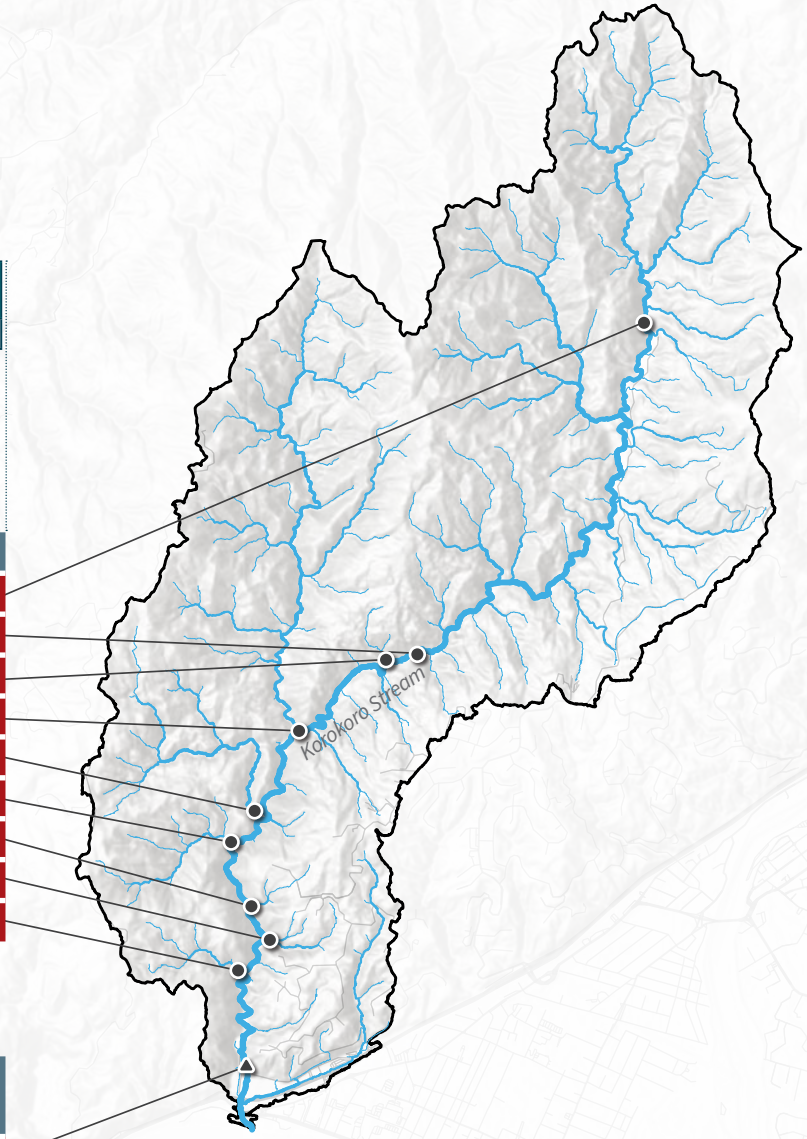
Rainfall: Korokoro Stream at Belmont Trig	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs
Rainfall recorded in mm	0	0	0	0	0.8	0.8	2.8	2.8



KEG Site #	01/07/25	21/09/25	06/11/25	15/02/26
CBKØ9	0.0170	0.0138	0.0134	0.0141
CBKØ8	0.0148	0.0127	0.0121	0.0125
CBKØ7*	0.0192	0.0186	0.0192	0.0200
CBKØ6*	0.0136	0.0135	0.0134	0.0148
CBKØ5*	0.0207	0.0240	0.0230	0.0300
CBKØ4*	0.0234	0.0270	0.0270	0.0340
CBKØ3*	0.0226	0.0314	0.0300	0.0300
CBKØ2*	0.0544	0.0741	0.0720	0.0820
CBKØ1*	0.0252	0.0333	0.0330	0.0420

* Indicates site on tributary of Korokoro Stream.

GW Site #	3-year 10-month Median	01/07/25	21/09/25	06/11/25	15/02/26
RS68	0.015	0.0148	0.0108	0.0109	0.0154



Water quality:
Escherichia coli (*E. coli*)
(n/100 ml)

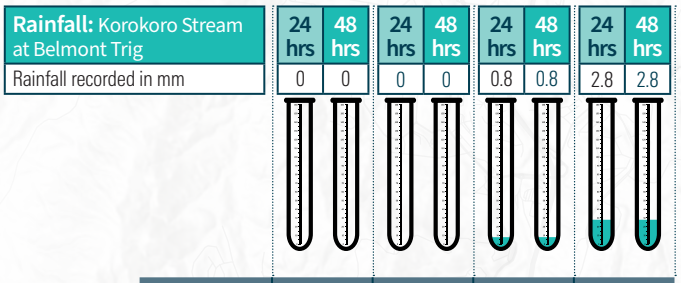
- *E. coli* levels were generally low across the stream, which indicates favourable conditions. Higher readings (at 'alert' or 'action' levels) were only recorded once, during the February 2026 sampling.
- The highest result was found in one tributary (CBKØ3), which has some urban areas upstream. This suggests that local land use may be affecting water quality at that site.
- Overall, the low *E. coli* levels are consistent with expectations for a predominantly forested catchment, with limited farming or urban development.
- It's also encouraging that *E. coli* levels at the GW SOE monitoring site (Cornish Street Bridge) are low compared to some other streams in the region, such as the Kaiwharawhara and Waiwhetu Streams.

GW LONGER-TERM DATA FROM NEARBY SITES FOR COMPARISON – from www.lawa.org.nz	
GW SOE Monitoring Site	**5-year Median *E. coli***
Hutt River at Boulcott	140
Waiwhetu Stream at Whites Line East	1,000
Parkvale Stream at Weir	295
Kaiwharawhara Stream at Ngaio Gorge	1,600

Water quality: *E. coli* (n/100 ml)

MfE/MOH (2003) surveillance, alert, & action levels		
Surveillance	Alert	Action

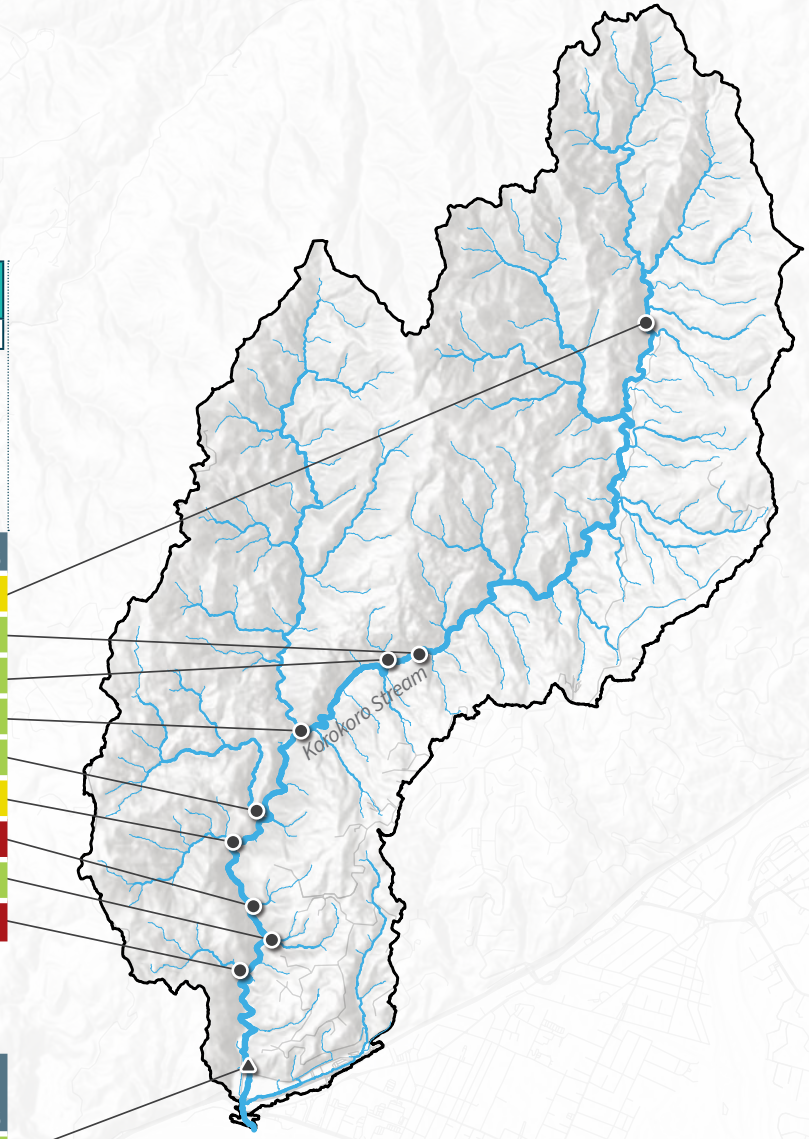
Rainfall: Korokoro Stream at Belmont Trig	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs
Rainfall recorded in mm	0	0	0	0	0.8	0.8	2.8	2.8



KEG Site #	01/07/25	21/09/25	06/11/25	15/02/26
CBKØ9	10	1	13	510
CBKØ8	10	8	21	52
CBKØ7*	1	1	2	3
CBKØ6*	15	1	2	13
CBKØ5*	9	4	4	25
CBKØ4*	47	6	19	320
CBKØ3*	1	1	1	1,500
CBKØ2*	16	2	4	120
CBKØ1*	120	1	10	650

* Indicates site on tributary of Korokoro Stream.

GW Site #	3-year 10-month Median	01/07/25	21/09/25	06/11/25	15/02/26
RS68	34	21	60	17	170



Water quality:
Nitrate-nitrogen (Nitrate-N) (mg/L)

- Nitrate-N levels were higher than recommended at almost all the monitoring sites. The only exception was one fully forested tributary (CBKØ7), which had much lower levels.
- Some tributaries had especially high nitrate-N levels, in particular CBKØ5 (which is also fully forested) and CBKØ2 (which has some urban areas upstream). Looking along the main stem itself, nitrate-N levels were fairly steady from upstream to downstream, with no clear pattern of increase or decrease.
- Importantly, none of the nitrate-N levels were high enough to be considered toxic to freshwater life.
- When compared with other streams in the region, nitrate-N levels in Korokoro Stream were generally higher than those in the Hutt River at Boulcott, but lower than those in streams with more intensive agricultural land use, such as Parkvale Stream in the Wairarapa

GW LONGER-TERM DATA FROM NEARBY SITES FOR COMPARISON – from www.lawa.org.nz	
GW SOE Monitoring Site	**5-year Median Nitrate-N**
Hutt River at Boulcott	0.166
Waiwhetu Stream at Whites Line East	0.475
Parkvale Stream at Weir	1.41
Kaiwharawhara Stream at Ngaio Gorge	0.96

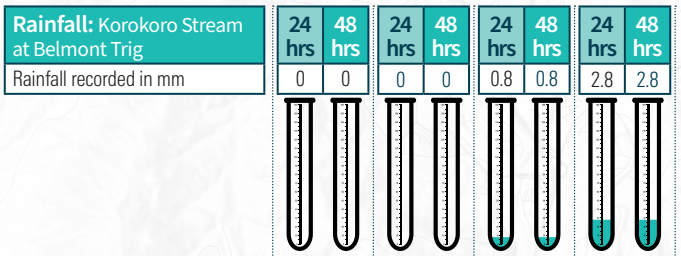
Water quality: Nitrate-N (mg/L)

ANZG (2018) DGV = 0.170
(whole catchment the same – based on Cool-Wet, Low Elevation REC climate and topography classes)

Indicates sample **OVER** DGV	Indicates sample **UNDER** DGV

GW NRP Toxicant Standard for nitrate-n = 1.5

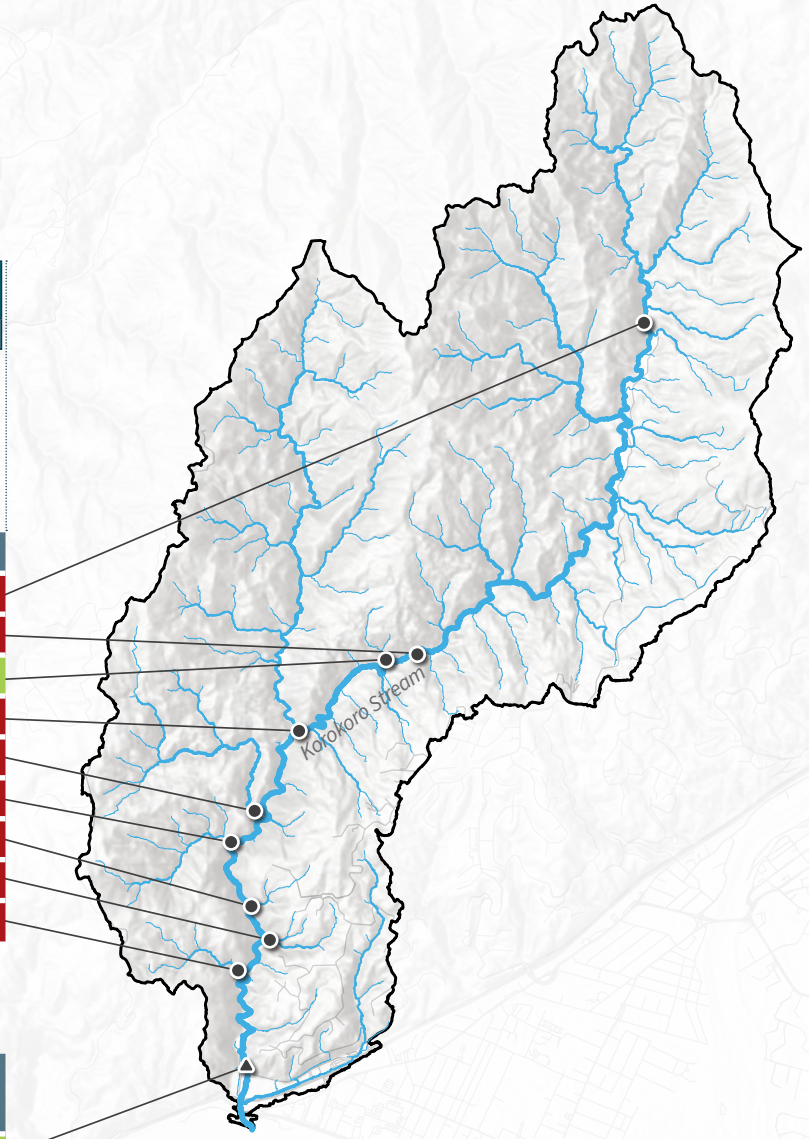
Rainfall: Korokoro Stream at Belmont Trig	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs
Rainfall recorded in mm	0	0	0	0	0.8	0.8	2.8	2.8



KEG Site #	01/07/25	21/09/25	06/11/25	15/02/26
CBKØ9	0.783	0.432	0.370	0.200
CBKØ8	0.676	0.460	0.380	0.210
CBKØ7*	0.062	0.0644	0.055	0.066
CBKØ6*	0.622	0.408	0.400	0.260
CBKØ5*	1.214	0.974	0.920	0.650
CBKØ4*	1.098	0.761	0.720	0.610
CBKØ3*	0.925	0.902	0.780	0.430
CBKØ2*	1.310	1.157	1.150	0.880
CBKØ1*	0.779	0.663	0.590	0.320

* Indicates site on tributary of Korokoro Stream.

GW Site #	3-year 10-month Median	01/07/25	21/09/25	06/11/25	15/02/26
RS68	0.40	0.706	0.413	0.340	0.148



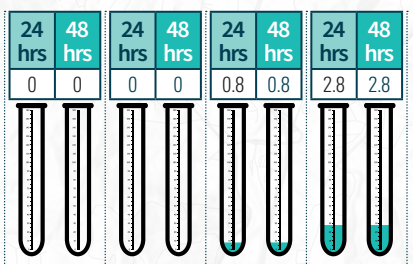
Water quality: Dissolved zinc (ug/L)

- Zinc was not detected at most of the monitoring sites. It was only found at a few tributary sites (CBKØ3, CBKØ2, and CBKØ1), and even then, the levels were very low and well within safe limits.
- These sites are in areas with some urban land use, so the small amounts of zinc detected are likely coming from things like roads or roofs.
- Overall, zinc is not a major concern in the Korokoro Stream based on the available data.

NRP Toxicant Standards (NRP-TS) **	
Indicates sample **OVER** standard	Indicates sample **UNDER** standard
BD = below detection	

* Values have been modified based on the mean hardness value for each site – see NRP TS column in tables below for site-specific standards.

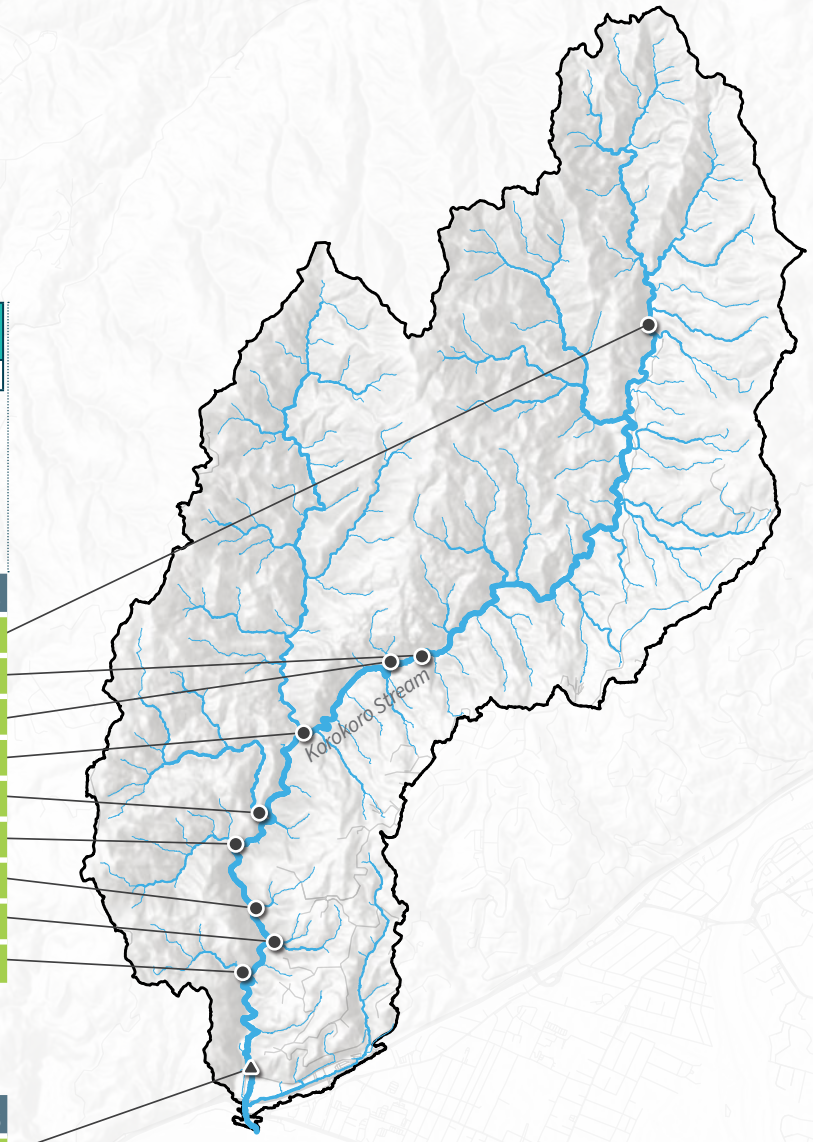
Rainfall: Korokoro Stream at Belmont Trig	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs
Rainfall recorded in mm	0	0	0	0	0.8	0.8	2.8	2.8



KEG Site #	NRP-TS	01/07/25	21/09/25	06/11/25	15/02/26
CBKØ9	8.05	BD (<1)	BD (<1)	BD (<1)	BD (<1)
CBKØ8	8.79	BD (<1)	BD (<1)	BD (<1)	BD (<1)
CBKØ7**	8.34	BD (<1)	BD (<1)	BD (<1)	BD (<1)
CBKØ6**	10.31	BD (<1)	BD (<1)	BD (<1)	BD (<1)
CBKØ5**	9.99	BD (<1)	BD (<1)	BD (<1)	BD (<1)
CBKØ4**	14.49	BD (<1)	BD (<1)	BD (<1)	BD (<1)
CBKØ3**	11.70	1.4	BD (<1)	BD (<1)	1.2
CBKØ2**	13.40	1.11	BD (<1)	1.2	1
CBKØ1**	12.66	BD (<1)	BD (<1)	BD (<1)	1.2

** Indicates site on tributary of Korokoro Stream.

GW Site #	NRP-TS	01/07/25	21/09/25	06/11/25	15/02/26
RS68	9.67	BD (<1)	BD (<1)	BD (<1)	BD (<1)

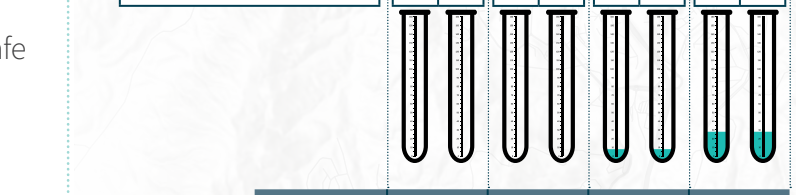


Water quality: Dissolved copper (ug/L)

- Copper was not detected at most of the sites. It was only found at two tributary sites (CBKØ2 and CBKØ1), and even there the levels were very low and well within safe limits.
- The small amount of copper found at CBKØ2 is likely linked to some urban land use in that part of the catchment.
- Overall, copper is not of concern for Korokoro Stream based on the available data.

NRP Toxicant Standards = 1.4 (whole catchment the same)	
Indicates sample **OVER** standard	Indicates sample **UNDER** standard
BD = below detection	

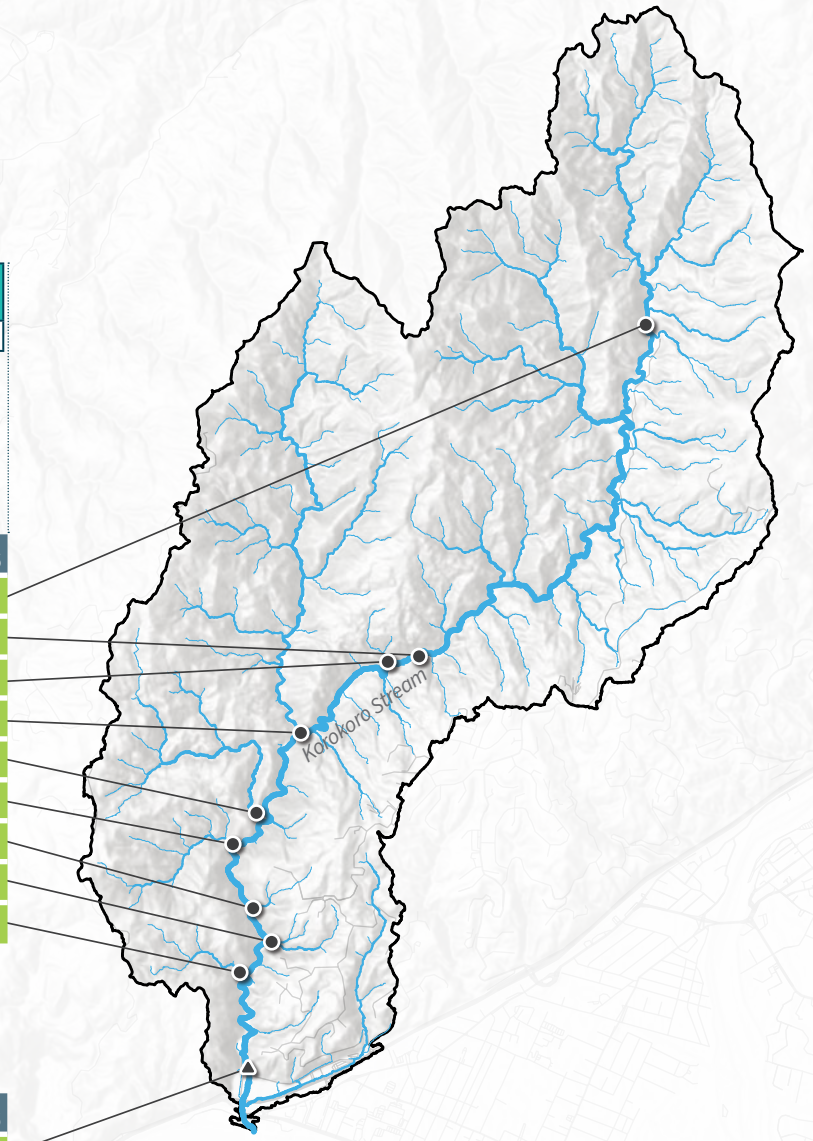
Rainfall: Korokoro Stream at Belmont Trig	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs	24 hrs	48 hrs
Rainfall recorded in mm	0	0	0	0	0.8	0.8	2.8	2.8



KEG Site #	01/07/25	21/09/25	06/11/25	15/02/26
CBKØ9	BD (<0.5)	BD (<0.5)	BD (<0.5)	BD (<0.5)
CBKØ8	BD (<0.5)	BD (<0.5)	BD (<0.5)	BD (<0.5)
CBKØ7*	BD (<0.5)	BD (<0.5)	BD (<0.5)	BD (<0.5)
CBKØ6*	BD (<0.5)	BD (<0.5)	BD (<0.5)	BD (<0.5)
CBKØ5*	BD (<0.5)	BD (<0.5)	BD (<0.5)	BD (<0.5)
CBKØ4*	BD (<0.5)	BD (<0.5)	BD (<0.5)	BD (<0.5)
CBKØ3*	BD (<0.5)	BD (<0.5)	BD (<0.5)	BD (<0.5)
CBKØ2*	0.59	BD (<0.5)	0.5	0.6
CBKØ1*	BD (<0.5)	BD (<0.5)	BD (<0.5)	0.6

* Indicates site on tributary of Korokoro Stream.

GW Site #	01/07/25	21/09/25	06/11/25	15/02/26
RS68	BD (<0.5)	BD (<0.5)	BD (<0.5)	BD (<0.5)



Macroinvertebrates

We don't have much information yet about the small animals that live in the stream – like insects and other benthic invertebrates (small, spineless creatures that live on/in sediment at the bottom of rivers, lakes, streams, and oceans).

So far, KEG have only taken kick net samples at two locations: 1) near the bottom of the catchment at Cornish Street Bridge, and 2) further upstream above the Woollen Mills Dam (which was sampled just once in 2019).

In addition to kick net sampling, taking environmental DNA (eDNA) samples helps identify species present by detecting traces of their DNA in the water. See Appendix 1 for more information about eDNA.

These two methods work in completely different ways. Kick net sampling collects animals directly from a small area of the streambed, while eDNA can pick up traces of species from much further upstream (sometimes hundreds of metres or even kilometres away). Because of this, the results from these two methods can't be directly compared.

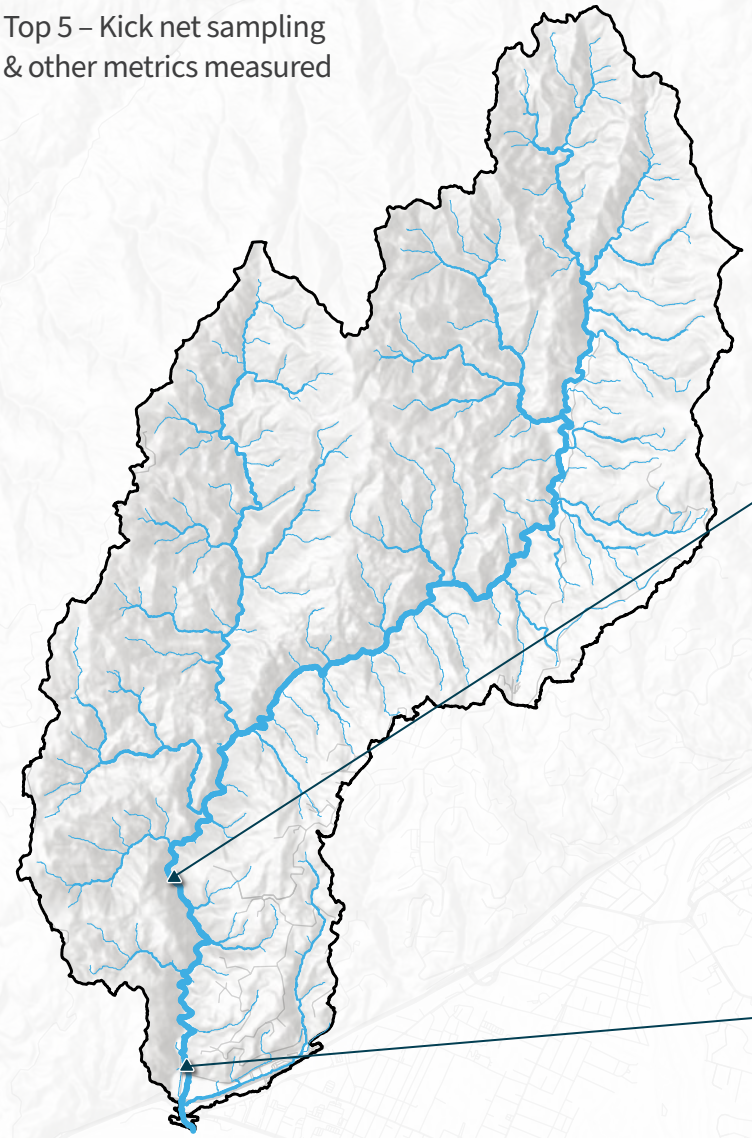
Kick net sampling

At the upstream site (RAN114), which was sampled once in 2019, the stream was in excellent condition and close to what you'd expect in a very natural, undisturbed environment. A range of sensitive stream insects were found there, including mayflies, stoneflies, and caddisflies. Most of the animals collected were just a few common types, particularly *Deleatidium* mayflies and two types of caddisflies (*Hydropsyche* and *Pycnocentrodes*). Together, these made up about 80% of what was found. These insects are usually a good sign of clean, healthy water, although some can still survive in streams that are less healthy.

At the downstream site at Cornish Street Bridge (RS68), conditions were not quite as good, but the stream is still in reasonably healthy shape overall. This site also had a variety of mayflies, stoneflies, and caddisflies. However, compared to the upstream site, there were more midge and crane fly larvae. These types of insects tend to become more common in streams that have more algae or are under some environmental pressure. This may be because the downstream site is more open and gets more sunlight, which can encourage algal growth that these insects feed on. It's important to keep in mind that, because we only have limited data, these comparisons should be treated carefully.

To better understand the health of the whole catchment, more sampling is needed, especially in the upper parts of the stream and its smaller tributaries, where there is currently little data. Regular yearly monitoring at the Cornish Street Bridge site (RS68) will also help track how conditions are changing over time, particularly in the lower catchment.

Macroinvertebrates:
Top 5 – Kick net sampling
& other metrics measured



GW Site #	Top 5 Most Abundant Taxa Based on Relative Abundance – KICK NET SAMPLING				
	1st	2nd	3rd	4th	5th
RAN114					
	Deleatidium mayflies (38%)	*Hydropsyche* caddisflies (21%)	*Pycnocentrodes* caddisflies (21%)	*Archichauliodes* dobsonflies (6%)	*Elmidae* beetles (6%)

METRICS						
Data Date	Length of Record	MCI	QMCI	ASPM	%EPT	Taxa Richness
April 2019	One-off	127.5 (Excellent)	6.17 (Excellent)	0.61 (Excellent)	83.90	16

GW Site #	Top 5 Most Abundant Taxa Based on Relative Abundance – KICK NET SAMPLING				
	1st	2nd	3rd	4th	5th
RS68					
	Hydropsyche caddisflies (19%)	Orthocladiinae midge larvae (15%)	*Aphrophila* cranefly larvae (15%)	*Maoridiamesa* midge larvae (11%)	*Archichauliodes* dobsonflies (6%)

METRICS						
Data Date	Length of Record	MCI	QMCI	ASPM	%EPT	Taxa Richness
Dec 2024	Annually since summer 2022–23 (3 years)	112.6 (Good)	4.66 (Fair)	0.55 (Good)	59.31	28

All species images © EOS Ecology unless otherwise specified



Kick net sampling. Image © EOS Ecology



Gillian eDNA sampling. Image © Greater Wellington

Environmental DNA (eDNA) sampling

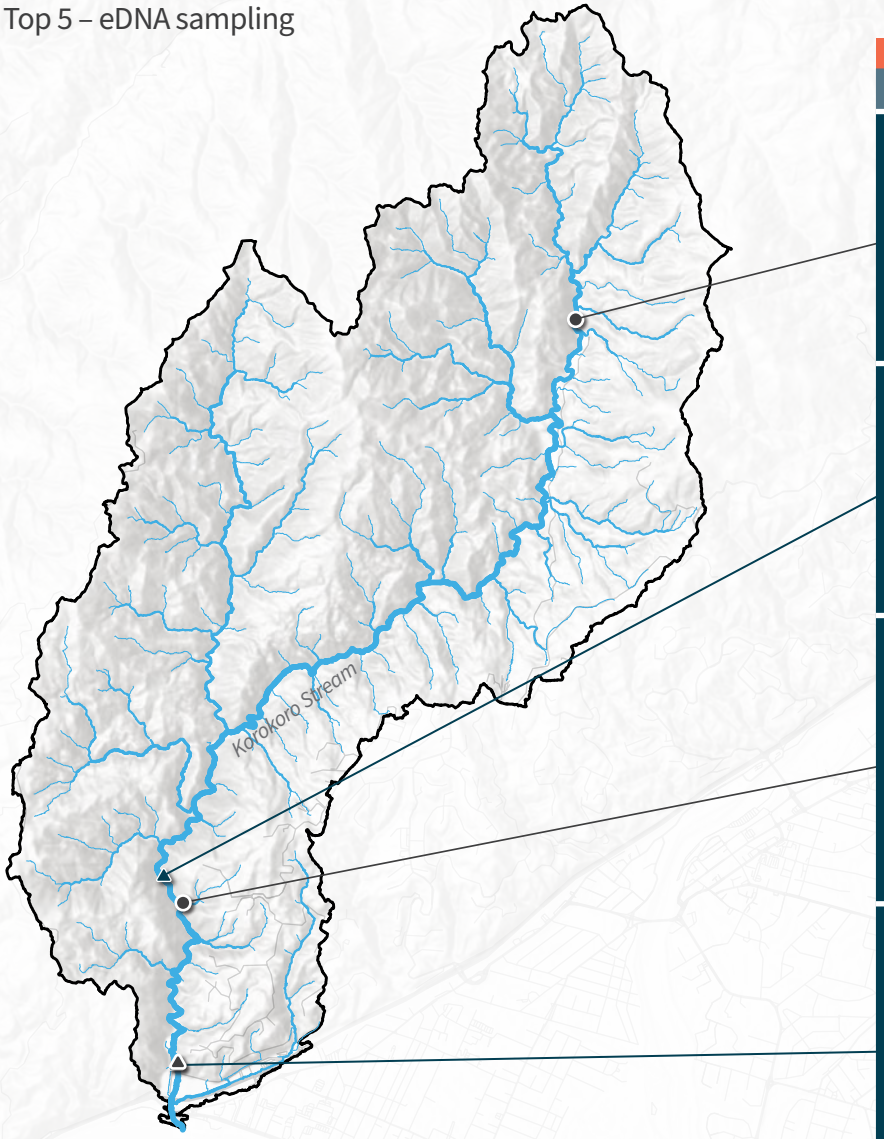
Environmental DNA results showed that one unexpected species, an exotic freshwater jellyfish (*Craspedacusta sowerbii)*, was frequently detected. It had the highest DNA signal at two sites (CBKØ3 and RS68) and was also common at another site (KOR0Ø2). This species is rarely found using traditional sampling methods but is often detected with eDNA.

Worm species were also very common, with four types appearing among the top results at several sites (CBKØ9, CBKØ3, and KOROØ2). Some more sensitive insects that are usually linked to healthier streams, such as *Coloburiscus humeralis* mayflies and *Hydrobiosis parumbripennis* caddisflies were also detected, but less consistently.

It's important to note that some sites were sampled more intensively than others. For example, six samples were taken at RS68 and KOROØ2, but only one sample was taken at CBKØ9 and CBKØ3. Because of this, it's not possible to fairly compare all sites directly.

When comparing the two most well sampled sites, many more sensitive insect species were found at the upper stream site (KOROØ2, with 36 species) than at the lower site (RS68, with 10 species). This could be because the upper site is less affected by urban runoff. However, more repeated sampling would be needed to confirm whether this difference is real or just due to natural variation in the data.

Macroinvertebrates:
Top 5 – eDNA sampling



Top 5 Taxa Based on DNA Sequence Count Average – eDNA SAMPLING					
Site #	1st	2nd	3rd	4th	5th
CBKØ9					
	Coloburiscus humeralis mayflies (410)	*Lumbriculus variegatus* blackworm (298)	*Eiseniella tetraedra* squaretail worm (169)	*Stylodrilus heringianus* worm (77)	*Potamopyrgus antipodarum* snail (67)
RAN114					
	Craspedacusta sowerbii freshwater jellyfish (4,706)	*Eiseniella tetraedra* squaretail worm (2,736)	*Lumbriculus variegatus* blackworm (2,636)	*Nais communis/ variabilis* complex sp. A1 (1,822)	*Archichauliodes diversus* dobsonflies (1,242)
CBKØ3					
	Chydorus sphaericus complex sp. A1 'water flea' cladoceran (5,845)	*Craspedacusta sowerbii* freshwater jellyfish (3,855)	*Lumbriculus variegatus* blackworm (3,180)	*Nais communis/ variabilis* complex sp. A1 (3,033)	*Coloburiscus humeralis* mayflies (2,359)
RS68					
	Craspedacusta sowerbii freshwater jellyfish (1,680)	*Oxyethira albiceps* caddisflies (424)	*Astrohydra japonica* hydra (368)	*Hydrobiosis parumbripennis* caddisflies (148)	*Tanytarsus* sp. EJD-2015 non-biting midge larvae (144)

All species images © EOS Ecology unless otherwise specified

Macroinvertebrates: Mayfly, stonefly and caddisfly species, and unique genera identified from kick net samples and eDNA samples

MAYFLIES (EPHEMEROPTERA)						
	RS68		RAN114	CBKØ3	KOROØ2	CBKØ9
Species	Kick Net	eDNA (6 reps)	Kick Net	eDNA (1 rep)	eDNA (6 reps)	eDNA (1 rep)
Austroclima spp.	✔		✔			
Deleatidium spp.	✔		✔			
Nesameletus spp.			✔			
Coloburiscus spp.			✔			
Zephlebia spp.	✔					
Deleatidium lillii		✔		✔	✔	✔
Deleatidium myzobranchia		✔		✔	✔	
Coloburiscus humeralis				✔	✔	✔
Ameletopsis perscitus				✔	✔	
Ichthybotus hudsoni				✔	✔	
Zephlebia borealis				✔	✔	
Nesameletus wainuimetea					✔	
Nesameletus ornatus					✔	
Nesameletus flavitinctus					✔	
Austroclima sepia					✔	
Neozephlebia scita					✔	
Acanthophlebia cruentata					✔	
STONEFLIES (PLECOPTERA)						
Zelandoperla spp.	✔		✔			
Spaniocerca spp.	✔					
Zelandobius spp.	✔					
Spaniocerca zelandica				✔	✔	
Acroperla trivacuata				✔	✔	
Zelandoperla decorata				✔	✔	
Zelandobius confusus					✔	

...table continues next column...

CADDISFLIES (TRICHOPTERA)						
	RS68		RAN114	CBKØ3	KOROØ2	CBKØ9
Species	Kick Net	eDNA (6 reps)	Kick Net	eDNA (1 rep)	eDNA (6 reps)	eDNA (1 rep)
Olinga spp.	✔		✔			
Pycnocentrodes spp.			✔			
Hydropsyche spp.	✔		✔			
Diplectrona spp.						✔
Costachorema spp.	✔		✔			
Hydrobiosis spp.	✔		✔			
Neurochorema spp.	✔		✔			
Psilochorema spp.	✔					
Plectrocnemia spp.	✔					
Pycnocentria spp.	✔					
Oxyethira albiceps		✔		✔	✔	
Psilochorema mimicum					✔	✔
Psilochorema donaldsoni		✔		✔	✔	
Hydrobiosis parumbripennis		✔		✔	✔	✔
Hydrobiosis gollanis				✔	✔	
Hydrobiosis soror				✔	✔	✔
Hydrobiosis copis		✔		✔		
Triplectides obsoletus		✔		✔	✔	
Triplectides dolichos					✔	
Hudsonema amabile						
Oeconesus maori					✔	✔
Olinga feredayi		✔		✔	✔	✔
Neurochorema confusum		✔			✔	
Costachorema xanthopterum		✔		✔	✔	
Polyplectropus aurifuscus					✔	
Beraeoptera roria					✔	
Hudsonema amabile					✔	
Pycnocentria evecta				✔	✔	
Pycnocentria funerea					✔	
Plectrocnemia maclachlani					✔	
Helicopsyche albescens					✔	
Zelolessica cheira					✔	
Total EPT taxa	**14**	**10**	**11**	**19**	**36**	**8**



Dobsonflies (*Archichauliodes diversus*). Image © EOS Ecology

Fish and kōura

Environmental DNA (eDNA) sampling

Most of the fish species identified through eDNA sampling were already known to live in the Korokoro Stream. The only exceptions were giant bully and shortjaw kōkopu. The shortjaw kōkopu finding is especially interesting, as this species is considered Threatened – Nationally Vulnerable and is often hard to detect using standard survey methods.

The eDNA results suggest that there may be fewer fish species further upstream. At the most upstream site (CBKØ9), only longfin eel and shortfin eel were detected. However, it's important to note that more samples were collected at some sites than others (especially at the downstream site, RS68), which likely influenced how many species were found in each location.

Longfin eel and shortfin eel were the only species detected at all sampling sites. Redfin bully and kōura (freshwater crayfish) were found at most sites, appearing at three out of the four locations sampled.

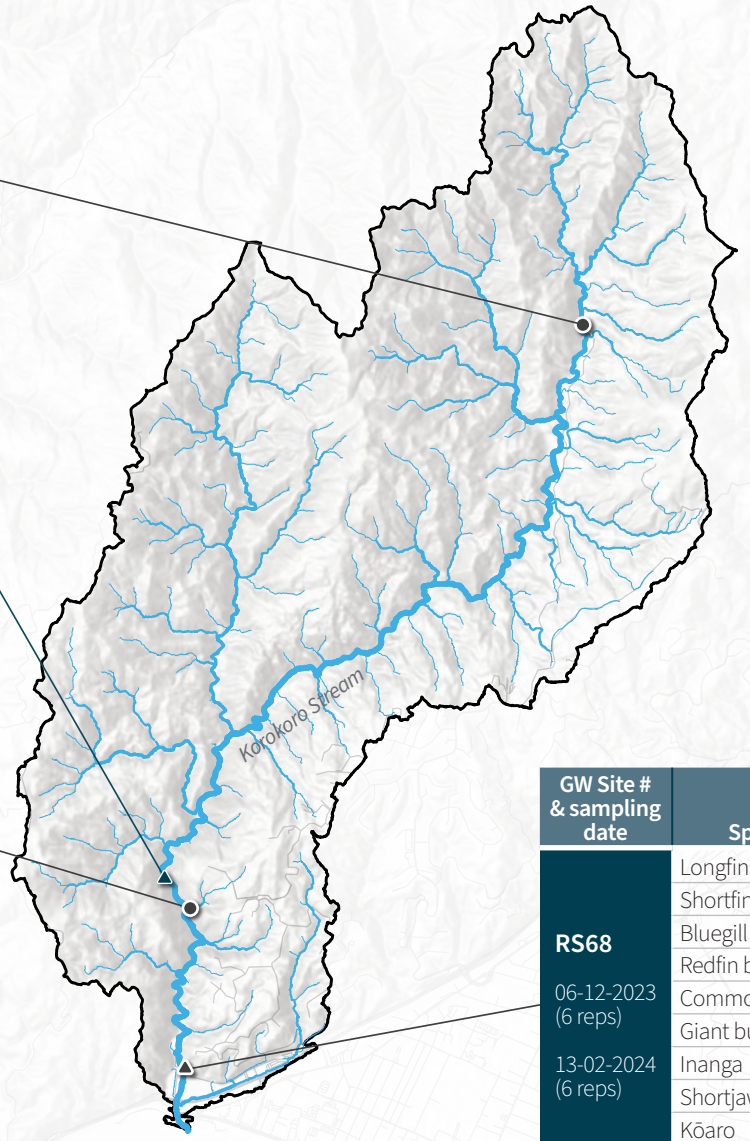
It's also worth keeping in mind that eDNA doesn't just reflect what's living exactly where the sample is taken. Because DNA can travel downstream in the water, the results may include species living further upstream, sometimes from hundreds of metres to several kilometres away. As a result, these findings give a broader picture of fish life across the stream, rather than just at each individual sampling spot.

Fish and Kōura: eDNA sampling

KEG Site # & sampling date	Species	Conservation Status
CBKØ9 28-06-2023 (1 rep)	Longfin eel	At Risk – Declining
	Shortfin eel	Not Threatened
	Kōura/freshwater crayfish	Not Threatened

GW Site # & sampling date	Species	Conservation Status
KOROØ2 11-01-2025 (6 reps)	Longfin eel	At Risk – Declining
	Shortfin eel	Not Threatened
	Redfin bully	At Risk – Naturally Uncommon
	Banded kōkopu	At Risk – Naturally Uncommon
	Kōaro	At Risk – Declining
	Kōura/freshwater crayfish	Not Threatened

KEG Site # & sampling date	Species	Conservation Status
CBKØ3 11-01-2025 (1 rep)	Longfin eel	At Risk – Declining
	Bluegill bully	At Risk – Naturally Uncommon
	Redfin bully	At Risk – Naturally Uncommon
	Common bully	Not Threatened
	Giant bully	At Risk – Naturally Uncommon
	Īnanga	Threatened – Nationally Vulnerable
	Brown trout	Introduced and Naturalised
	Kōura/freshwater crayfish	Not Threatened



GW Site # & sampling date	Species	Conservation Status
RS68 06-12-2023 (6 reps) 13-02-2024 (6 reps)	Longfin eel	At Risk – Declining
	Shortfin eel	Not Threatened
	Bluegill bully	At Risk – Naturally Uncommon
	Redfin bully	At Risk – Naturally Uncommon
	Common bully	Not Threatened
	Giant bully	At Risk – Naturally Uncommon
	Inanga	Threatened – Nationally Vulnerable
	Shortjaw kōkopu	Threatened – Nationally Vulnerable
	Kōaro	At Risk – Declining
	Brown trout	Introduced and Naturalised

5. Proposed Petone to Grenada highway

A new four-lane highway is proposed to connect Petone with northern Wellington/Porirua. Known as the Petone to Grenada (P2G) highway, as of June 2026 this project is in the consent investigations stage with the intent to lodge an application under the Fast-track Approvals Act (2024) in late 2026. The current preferred route of P2G traverses the lower part of the Korokoro Stream catchment, which has already been modified by commercial and industrial development. If P2G proceeds, this will present both risks and opportunities for the lower reaches of the Korokoro Stream.

Overall, the P2G would have some adverse effects on the lower part of the Korokoro Stream, but through effective management these should be minimised or avoided (e.g., strict erosion and sediment controls, stormwater treatment, ensuring fish passage). Most large roading infrastructure projects require some level of biodiversity offsetting for both terrestrial and freshwater habitats. There is the potential some of this work could be undertaken in the Korokoro Stream catchment and provide long lasting benefits.

Risks include:

- habitat disturbance and contamination from the demolition of existing commercial and industrial buildings
- inputs of fine sediments and other contaminants during construction-phase earthworks
- ongoing inputs of highway stormwater once the highway is operational
- permanent loss of regenerating native vegetation.

Opportunities include:

- removal of buildings that have been constructed across the stream and their associated old culverts, with replacement by a modern, fish friendly culvert solution
- construction of stormwater treatment infrastructure to capture contaminants prior to discharge to Korokoro Stream
- potential for some of the P2G's biodiversity offsetting requirements to be directed towards the Korokoro Stream catchment.

6. Next steps

Actions:

- Remain up-to-date with the P2G consenting process and advocating, where possible, for the ecological and community values of the Korokoro Stream catchment.
- If P2G does proceed, ensure some of the biodiversity offsetting requirements are directed at the Korokoro Stream catchment.
- Investigate further fish barrier improvements, where possible, especially for the Woollen Mills Dam and Korokoro Dam.
- Restore īnanga spawning habitat at stream mouth.

Monitoring:

- Further water quality monitoring, including targeting larger rain events that generate stormwater runoff.
- Monitoring the condition and effectiveness of the fish passage improvements made to the catchment.
- Further fish monitoring to validate and confirm the presence of shortjaw kōkopu (eDNA record) and dwarf galaxias (a single NZFFD record).

7. References

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Korokoro Environmental Group (KEG) local Playcentre visit. Image © Greater Wellington

Appendix 1: Environmental DNA (eDNA) explained



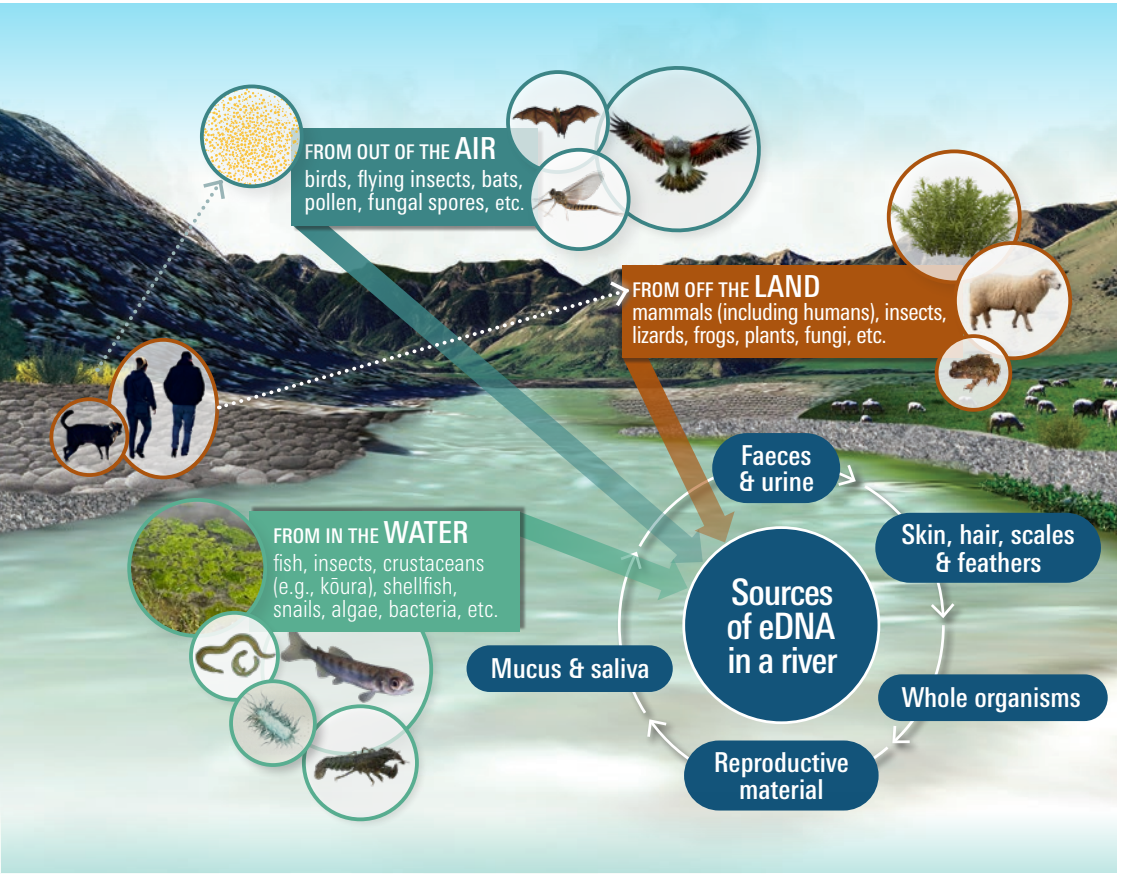
Environmental DNA (eDNA) sampling. Image © Greater Wellington

Sources of eDNA in waterways

The genetic information for every living organism is stored within a molecule called DNA (deoxyribonucleic acid). It acts as a blueprint for how the organism will look and function.

Each species has short DNA sequences unique to that species that can be used for identification. A full copy of an organism's DNA is in nearly every cell, which means a single cell can identify what species the organism belongs to – even if the cell is no longer attached to the organism, i.e., it comes from material that has dropped off or been released by an organism.

DNA shed by an organism into water, soil or air while going about its day is called environmental DNA, or eDNA. The infographic below illustrates this concept.

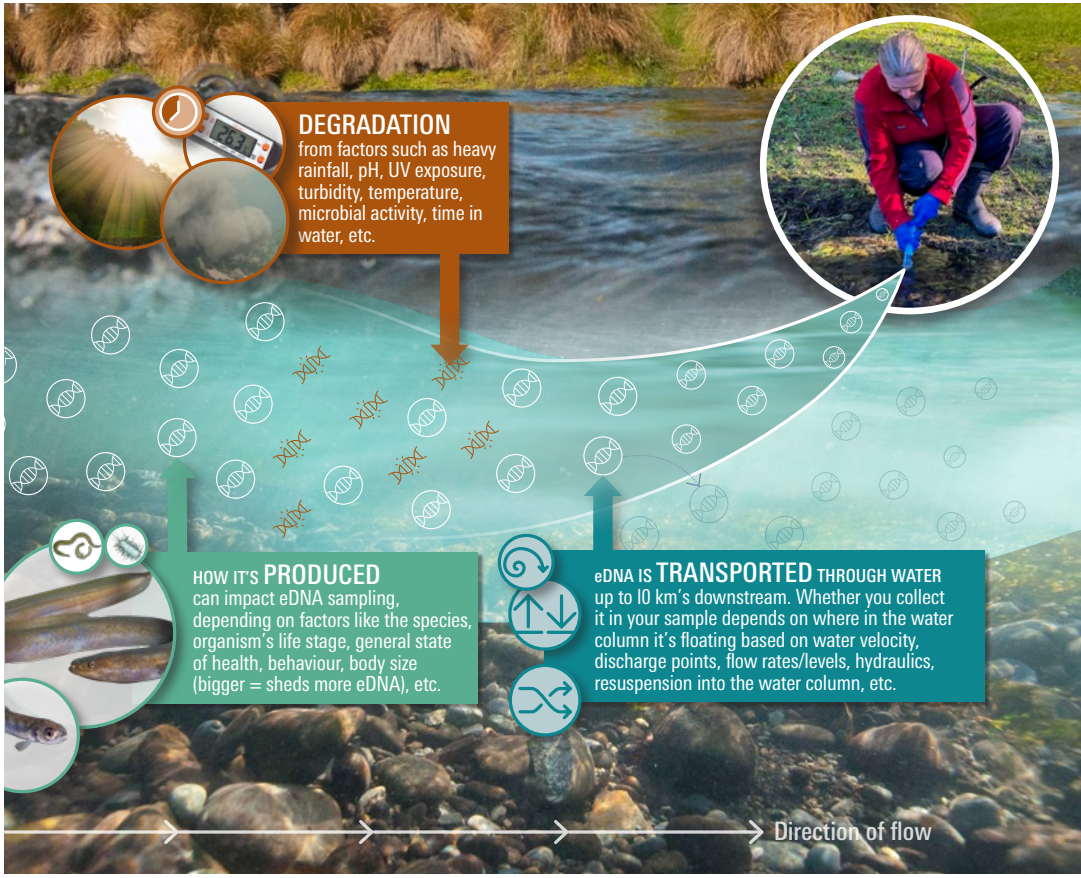


Factors affecting eDNA detectability

Some of the limitations with eDNA data include but are not limited to the following:

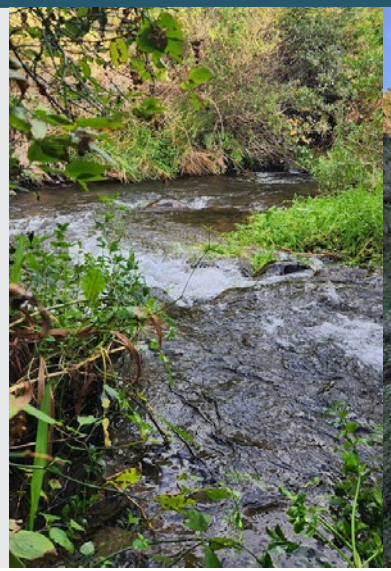
- Only detectable for a short period of time, and the time frame is influenced by environmental conditions (e.g., detection rates may decline after heavy rain or other environmental factors).
- Human influence (e.g., sewage discharges) can introduce foreign DNA, potentially leading to overestimating biodiversity, while fluctuations in water flow can transport eDNA from upstream – also influencing perceived species distribution.

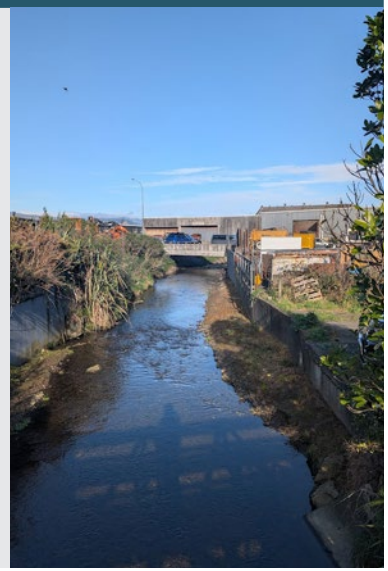
REMEMBER...eDNA sample results only indicate presence/absence of species within the sample you've taken. There's no correlation between the amount of eDNA found for an individual species with their abundance!



Infographics © EOS Ecology. Cannot be used without prior written permission.

Appendix 2: KEG sampling site photos & description

	CBKØ9: Korokoro Stream at Woolshed	**CBKØ8**: Korokoro Stream 50m below dam	**CBKØ7**: Korokoro Stream trib 250m below Korokoro Dam	**CBKØ6**: Korokoro Stream trib at Korokoro Forks	**CBKØ5**: Korokoro Stream trib west of bridge
Site					
Description	Mainstem of the upper Korokoro stream. Site mixed native vegetation and maintained lawn.	Mainstem of the Korokoro stream just below the Korokoro dam. Forested site in native bush.	Small, steep tributary on the true left bank. Located just below the Korokoro dam.	Large tributary on the true right bank. Drains the northwest part of Belmont Regional Park.	Small tributary on the true right bank. Flows under a small bridge into the stream.

	CBKØ4: Korokoro Stream trib 400m above Woollen Mills Dam	CBKØ3: Korokoro Stream trib below Titiro Moana Road	CBKØ2: Korokoro Stream trib below Gatehouse Lane	CBKØ1: Korokoro Stream trib 900m above Cornish St bridge	RS68: Korokoro Stream at Cornish Street Bridge
Site					
Description	Tributary on the true right bank. Accessible only from crossing the Korokoro Stream.	Small, steep tributary on the true left bank, with waterfall above. Flows through a drainage pipe across the trail and into the stream. Located just below the Woollen Mills dam.	Steep tributary on the true left bank, flows under a small bridge into the stream.	Small tributary on the true right bank that flows under a small bridge into the stream.	SOE site monitored since July 2022. This site represents the lowest site accessible before the stream enters a piped system and flows into the Wellington harbour. It is located in an industrial area.

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